Delaram Fartoot
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Profile

AI/ML Engineer with experience building **LLM and retrieval-based systems**, specializing in **RAG, information retrieval, and embedding models**. Experienced in designing modular RAG pipelines, benchmarking state-of-the-art retrieval models on **MTEB and Persian-domain FamTEB**, and researching retrieval architecture improvements. Strong Python and machine learning background with hands-on experience in **LangChain, Docker, MLflow, DVC, PostgreSQL**, and ML experimentation workflows.

Education

- **University of Tehran** — Tehran, Iran
  Bachelor of Computer Science; GPA: 18.55/20.00 — *Sep. 2020 – Feb. 2025*
- **National Organization for Development of Exceptional Talents (Farzanegan 1)** — Tehran, Iran
  Diploma in Mathematics and Physics; GPA: 19.92/20.00 — *Sep. 2017 – Sep. 2020*

Experience

- **Pars Iranian Artificial Intelligence Technologies Company (website)** — Tehran, Iran
  Machine Learning Researcher — *May. 2025 – Present*
  - Designed and implemented a **modular retrieval pipeline using LangChain**, developing configurable components for chunking, indexing, retrieval, and search; achieved **82.42 NDCG@5** on the company's internal dataset.
  - Researched and benchmarked **state-of-the-art retrieval and RAG approaches**, including hybrid RAG, reranking, reliable RAG, and multimodal RAG; evaluated a wide range of embedding and retrieval models on **MTEB and Persian-domain FamTEB** benchmarks.
  - Investigated modifications to **positional encoding mechanisms**, including **RoPE**, within retrieval model architectures, resulting in improved retrieval performance.
  - Worked with modern **ML engineering and MLOps tools**, including **Docker, MLflow, DVC, and PostgreSQL**, gaining hands-on experience with containerization, experiment tracking, dataset/version management, and ML data infrastructure.

- **Heinrich Heine University Düsseldorf** — Düsseldorf, Germany
  Research Assistant Intern under supervision of Prof. Swoboda — *Jul. 2024 – Sep. 2024*
  - Implemented decision diagrams for various problems, including the knapsack problem, to represent and solve complex decision-making scenarios.
  - Customized Gym environments to integrate **Proximal Policy Optimization (PPO)** and reinforcement learning, achieving optimal decision diagram approximations.

- **IPM Institute for Research in Fundamental Sciences** — Tehran, Iran
  Research Assistant Intern under supervision of Prof. Rahmati — *Jul. 2022 – Sep. 2023*
  - Conducted research on reducing bias in NLP by enhancing classical models with **LLM paraphrasers**, improving fairness while preserving F1 scores for personality trait classification from social media data.
  - Assembled a **Persian-language medical question-answering dataset**, addressing resource gaps in Persian NLP and exploring applications in healthcare AI.

- **University of Tehran** — Tehran, Iran
  Teaching Assistant — *Jan. 2023 – Jul. 2024*
  - Teaching Assistant for **Data Mining**.
  - Head Teaching Assistant for Differential Equations.
  - Teaching Assistant for Foundations of Combinatorics.

Skills

- **Programming Languages**: Python, C++, SQL, R, MATLAB, JavaScript
- **Machine Learning & AI**: PyTorch, TensorFlow, Scikit-learn, Hugging Face, Fine-tuning, Representation Learning
- **Data & Engineering**: Pandas, NumPy, PostgreSQL, REST APIs, Git, Linux
- **MLOps & Infrastructure**: Docker, MLflow, DVC, ETL/ELT Concepts, Weights & Biases
- **Frameworks & Tools**: LangChain, Gymnasium, Django
- **Languages**: Persian (Native), English (**IELTS Overall: 8.0**), German (Familiar)

Selected Projects

Network Intrusion Detection *(code)*

- Developed a network intrusion detection system utilizing various machine learning models, including **stacking**, achieving an F1 score of **81.18%**, comparable to results in the referenced research paper.

Persian Author Identification *(code)*

- Developed and evaluated **BERT and SVM** models to classify Persian authors from different historical periods based on their works and stylistic features.

Medical Specialty Diagnosis

- Developed a machine learning system for medical specialty diagnosis from clinical transcriptions, achieving **94% accuracy** across three specialties.

MBTI Classification

- Designed a machine learning model to classify MBTI personality traits from tweets, addressing challenges posed by highly imbalanced data.

Achievements

- Awarded a scholarship at the **Dialog Systems and Machine Learning Chair**, Heinrich Heine University Düsseldorf.
- Achieved academic excellence in **8 of 9 semesters** at the University of Tehran.
- Ranked in the **top 0.2%** in Iran's National Universities Entrance Examination.

Relevant Coursework

- Data Mining (20/20)
- Artificial Intelligence (19/20)
- Advanced Information Retrieval (20/20)
- Database (20/20)
- Principles of Software Design (18.5/20)
- Principles of Operating Systems (20/20)
- Design and Analysis of Algorithms (20/20)
- Data Structures and Algorithms (18/.21/20)

Courses & Certifications

- **Stanford Online**: CS229 Machine Learning, CS224N Natural Language Processing with Deep Learning, CS234 Reinforcement Learning.
- **Coursera**: Natural Language Processing with Classification and Vector Spaces, Natural Language Processing with Probabilistic Models, Supervised Machine Learning: Regression and Classification, Advanced Learning Algorithms.